

Introduction to the Theory of Computation

Lecture 6. Nonregular Languages and the Pumping Lemma

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Given a computation model, we want to understand its **computability**: which languages are computable, and which are not?

For finite automata, only **regular languages** are computable.

Almost All Languages Are Nonregular

Theorem 1

Almost all languages are nonregular.

Proof.

Every regular language is described by some DFA, and each DFA can be encoded as a finite string. Since there are only **countably many** such strings, there are only countably many regular languages.

The set of all languages is $\mathcal{P}(\Sigma^*)$, and since Σ^* is countably infinite,

$$|\mathcal{P}(\Sigma^*)| = 2^{\aleph_0},$$

which is uncountable (by Cantor's theorem). Since $\aleph_0 < 2^{\aleph_0}$, almost all languages are nonregular. □

How Do We Prove a Language Is Not Regular?

We know almost all languages are nonregular.
But how do we prove a **specific** language is nonregular?

⇒ The **Pumping Lemma**.

The Pumping Lemma

Lemma 2 (Pumping Lemma)

If L is a regular language, then there exists $p \in \mathbb{N}$ such that, for any string $s \in L$ with $|s| \geq p$, we can write

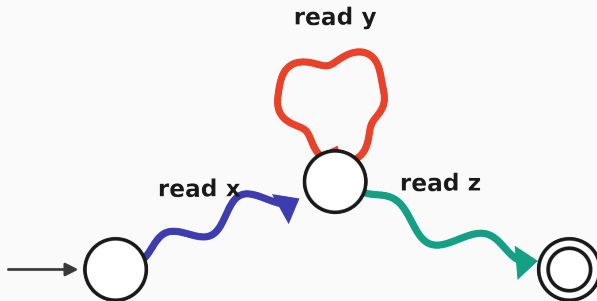
$$s = xyz, \quad x, y, z \in \Sigma^*,$$

such that

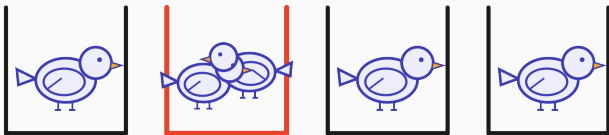
1. $xy^iz \in L \quad \forall i \geq 0$,
2. $|y| > 0$, and
3. $|xy| \leq p$.

Proof Idea

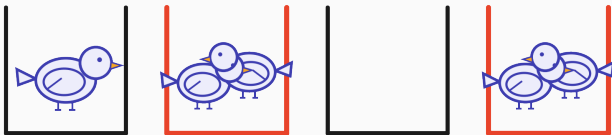
Let $p = |Q|$. Since $|s| \geq p$, the run visits at least $p + 1$ states. By the **pigeonhole principle**, some state is visited more than once.



Pigeonhole Principle



Pigeonhole Principle



Proof of the Pumping Lemma (1/2)

Proof.

Let $M = (Q, \Sigma, \delta, q_0, F)$ be a DFA recognizing L . Let $p = |Q|$. Let $s = s_1s_2 \cdots s_n \in L$ be a string of length n , where $n \geq p$. Let $r_0, r_1, \dots, r_n \in Q$ be the sequence of states visited on input s , i.e.

$$\begin{aligned} r_0 &= q_0, \\ r_i &= \delta(r_{i-1}, s_i) \quad \text{for } 1 \leq i \leq n. \end{aligned}$$

By the pigeonhole principle, since there are $n + 1 > p$ states in this sequence but only p states in Q , some state must repeat.

Proof of the Pumping Lemma (2/2)

Let j be the smallest index such that $r_j = r_i$ for some $i < j$. Since $r_0, \dots, r_p$ are $p + 1$ states in Q , such a j exists and $j \leq p$. Let

$$x = s_1 \cdots s_i,$$

$$y = s_{i+1} \cdots s_j,$$

$$z = s_{j+1} \cdots s_n.$$

- Since $i < j$, $|y| \geq 1$.
- Since r_j is the first repeated occurrence, $j \leq p$, i.e. $|xy| \leq p$.
- Since $r_i = r_j$, the machine can loop through y any number of times $(0, 1, 2, \dots)$ and still end up at $r_n \in F$, so $xy^iz \in L$ for all $i \geq 0$.

This is exactly the idea behind proving a language is **not** regular: assume it is regular, then **pump** a long string and show some xy^iz falls outside the language — a contradiction.

The pumping lemma is a tool to **disprove** regularity.

Example: $\{0^n 1^n : n \geq 0\}$ Is Not Regular

Proof.

Suppose, for contradiction, that $L = \{0^n 1^n : n \geq 0\}$ is regular.

By the pumping lemma, there exists p such that the three conditions hold. Let $s = 0^p 1^p \in L$.

By the pumping lemma, $s = xyz$ with $|xy| \leq p$ and $|y| > 0$, which forces x, y to consist only of 0's. But then xy^2z has more 0's than 1's, so $xy^2z \notin L$ – contradicting $xy^i z \in L$ for all $i \geq 0$. Contradiction! □

In-Class Exercises: Prove Non-Regularity

Use the pumping lemma to show each of the following languages is not regular:

1. $\{0^n 1^{2n} : n \geq 0\}$
2. $\{w \in \{0, 1\}^* : w \text{ has an equal number of 0's and 1's}\}$
3. $\{ww : w \in \{0, 1\}^*\}$
4. Palindromes: $L = \{w \in \{0, 1\}^* : w^R = w\}$, where w^R is the reverse of w .
5. $\{0^n 1^m : n > m\}$
6. $\{0^n : n \text{ is a perfect square}\}$